

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: ESSENTIAL MATHEMATICS GRADE 10: 2.3 TRIGONOMETRY

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation. It answers the driving question: How can we use the angles and one side of a right-angled triangle to find any unknown height or distance in our environment, without ever measuring it directly? It also answers the key inquiry question: Why is trigonometry important? Write your answers in the space under each prompt. Use what you observed and learned in Lessons 1 to 8 and the Kilimani Secondary School water tank in Nakuru as your main example. Use a calculator in DEG mode. Give trig values to 4 decimal places and final lengths to 1 decimal place with units. Show every step of your working, draw labelled diagrams where asked, and use the words hypotenuse, opposite, adjacent, sin, cos and tan carefully.

Part 1: The triangle and the ratio that makes the method work

At Kilimani Secondary School nobody can climb the steel-framed water tank tower, but Wanjiru and Otieno can still find its height. (a) Draw and label the right-angled triangle formed by the tower, the level ground and the line of sight for Wanjiru, who stands 8 m from the base. Mark the angle θ at her eye, and label the hypotenuse, opposite and adjacent sides. Say what each side represents in the real situation. (b) Using the three 35° triangles A, B and C from Lesson 2 (sides 4.0, 2.8, 4.9 cm; 8.0, 5.6, 9.8 cm; 12.0, 8.4, 14.7 cm), explain what you found about the ratios when the triangles got bigger. (c) Explain why this idea matters for the tank.

Write your answer here:

Part 2: Choosing a ratio and calculating

(a) Explain how you decide whether to use sin, cos or tan when you need to find an unknown side. (b) Use the 3-step method (label, choose the ratio, solve with units) to find: (i) the length of a Kisumu roof rafter when the horizontal span from the wall to the centre is 3.6 m and the rafter makes 30° with the horizontal; (ii) the width of a Kericho stream, when a learner on the bank directly opposite a tree walks 20 m along the bank and the line of sight to the tree then makes 40° with the bank. (c) A Watamu diver's weighted line is 12 m long and makes 50° with the horizontal surface. Find the depth of the weight.

Write your answer here:

Part 3: Solving the water tank problem

Wanjiru stands 8 m from the base of the tower and reads an angle of elevation of 37° . Otieno stands 10 m from the base and reads 31° . Every learner's eye height is 1.5 m. (a) Explain how the clinometer gives the angle of elevation, and what the string does when the straw is level. (b) Write a complete solution for Wanjiru: the diagram description, the ratio and why you chose it, the calculation and the answer with units. (c) Do the same for Otieno. (d) Wanjiru is closer, so whose angle is larger? Explain why both learners should still get the same height.

Write your answer here:

Part 4: Checking the answer and sources of error

Group Amani used their clinometers on the school flagpole, standing 15 m from the base and reading 27° . Mr. Kamau, the school caretaker, used a tape to check the heights: the flagpole is 9.0 m and the top of the water tank is 7.5 m. (a) Calculate Group Amani's flagpole height. (b) Work out their error. (c) Explain why small differences like this happen. (d) Use your answers from Part 3 and Mr. Kamau's tape value for the tank to judge whether Wanjiru's and Otieno's answers are good.

Write your answer here:

Part 5: Answering the driving question

Write a clear explanation in your own words that answers: How can we use the angles and one side of a right-angled triangle to find an unknown height or distance in our environment, without measuring it directly? Use the Kilimani Secondary School water tank as your example. Include: the triangle, why the method works, the steps, and how you know the answer can be trusted. Then answer the key inquiry question: Why is trigonometry important? Give at least two real-life Kenyan examples.

Write your answer here:

RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Right-angled triangle model and side naming	Draws and labels the triangle correctly from θ , identifies the hypotenuse, opposite and adjacent sides, and links each to a real part of the tank situation.	Labels the triangle and sides mostly correctly, with a weak link to the real situation.	Labels are missing or incorrect, or opposite and adjacent are confused.
Understanding why ratios stay constant	Clearly explains that sin, cos and tan depend only on the angle, using the 35° triangle evidence, and links this to Wanjiru and Otieno getting the same height.	States that the ratios stay the same but gives limited evidence or a limited link to the tank.	Does not explain why the ratios stay constant, or says they change with triangle size.
Choosing the ratio and calculating	Chooses the correct ratio from the known and wanted sides with a reason, calculates accurately with the 1.5 m eye height where needed, and gives correct	Chooses the correct ratio and calculates mostly correctly, with a small slip in units, rounding or eye height.	Chooses the wrong ratio or makes major calculation errors.

	units and rounding.		
Checking results and explaining error	Compares calculated heights with the tape values, finds the small error of about 0.1 m and explains it by reading angles to the nearest degree.	Compares with the tape values but gives a limited explanation of the error.	Does not compare with the tape values or gives no valid reason for the error.
Answering the driving and key inquiry questions	Gives a complete, well-organised explanation of the method and why it works, and explains why trigonometry is important with two or more Kenyan examples.	Explains the method and gives one example of why trigonometry is important.	The explanation is incomplete or has no real-life examples.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

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Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

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